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Towards structure-property prediction in DyF_3 nanoclusters

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ABSTRACT

Rare-earth trifluoride nanoparticles attract increasing attention due to their potential use in biomedical applications and spintronics, where understanding the correlation between structure and magnetic properties is crucial. However, experimental determination of these properties at the nanoscale remains challenging. In this work, the structures and magnetic characteristics of $(DyF_3)_n$ (n = 1–10) nanoclusters were predicted and analyzed using an evolutionary algorithm combined with density functional theory. This study reveals how nanoclusters evolve structurally from symmetric monomers (C_{3v}) to less symmetric larger clusters (C_1), with Dy-F bonds (2.02–2.96 Å), Dy-Dy distances (3.43–8.42 Å), and coordination numbers increasing from 3 to 8. Notably, magnetic moments grow with size, peaking near $\approx 45\mu_B$ at $n = 10$, while spin compensation emerges at $n = 7$. Incorporating spin–orbit coupling highlights significant magnetic anisotropy energies up to 245 meV. Additionally, solvent effects provide substantial stabilization (−2.8 to −8.2 eV). These findings uniquely link structural changes to size-dependent magnetic properties, advancing predictive design of rare-earth fluoride nanomaterials.

1. Introduction

Rare-earth trifluorides (RETs) form a class of materials known for their outstanding electronic, optical, and magnetic properties. These attributes make them highly desirable for a wide range of technological and scientific applications, especially in their nanoparticle (NP) form. Nanoparticles are distinct from their bulk counterparts due to the size-dependent effects of surface phenomena and constrained geometries. These effects endow RET NPs with novel properties, enhancing their utility in biomedicine, catalysis, and optics [1,2].

RET NPs hold great potential as imaging contrast agents in biomedicine. Among them, dysprosium-based trifluoride NPs (DyF_3) have attracted interest for MRI applications due to their unique magnetic properties. Studies have shown size-dependent shifts in Curie temperature and magnetic ordering, highlighting their suitability for high-field imaging [3,4]. Other RET NPs, such as TbF_3 and HoF_3, have also been explored for dual-mode MRI/CT imaging, demonstrating low toxicity and high biocompatibility [5,6]. GdF_3-based particles further extend RET applications to multimodal imaging and drug delivery [7–9].

Beyond imaging, RET is used in medicine and biotechnology to create nanoprobes [10], where their biocompatibility, combined with unique magnetic and optical properties, makes them promising candidates for radiation therapy [11], photodynamic therapy [12], and nanothermometry [13].

The magnetic properties of RET are particularly significant when transitioning to the nanoscale. These materials exhibit diverse magnetic behaviors, especially at low temperatures, driven by dipole–dipole interactions. One of the key research questions involves understanding how the NP size influences the magnetic moment distribution, magnetic domain formation, and the critical size necessary for magnetic ordering. Surface effects add complexity to this investigation, as they substantially alter the magnetic behavior in core–shell and other NP geometries [14]. For example, studies on DyF_3 nanoparticles have revealed a pronounced size effect: magnetic ordering diminishes below

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a critical particle size, underscoring the intricate relationship between size and magnetic properties [15].

From a theoretical perspective, modeling the magnetic properties of rare-earth trifluoride nanoparticles poses significant challenges. Although density functional theory (DFT) provides a powerful tool for *ab initio* calculations, its application to systems with large atomic counts is computationally intensive. However, force-field-based methods cannot accurately simulate magnetic phenomena, resulting in a gap in understanding size-dependent magnetic behavior at the nanoscale. Notably, there are no examples of DFT simulations of RET nanoparticles, except our previous work [16]. In that study, spherical NPs were modeled by adapting an existing periodic crystal structure as a template. Although this approach provided valuable insights, it did not account for constructing NPs with fully relaxed ground-state energies, potentially limiting the accuracy of the predicted magnetic properties. To address these limitations, the present study adopts an alternative methodology, including evolutionary algorithms for constructing RET NPs, further described in detail in Section 2.

Besides RET, there are theoretical works predicting the magnetic properties of NPs. For example, Liu and Di Valentin explored the structural and magnetic properties of Fe_3O_4 NPs, providing insights into how their shape influences magnetic behavior [17]. Cuadrado and Chantrell modeled bimetallic $L1_0$ cuboctahedral FePt, CoPt, FeAu, and FePd clusters using fully relativistic DFT calculations to study the electronic and magnetic characteristics, focusing on size and structural influences [18]. Hu and coauthors examined the electronic, magnetic, and optical properties of transition metal-embedded ZnO nanostructures, highlighting the role of transition metals in modifying the magnetic properties of ZnO-based NPs [19]. Mai et al. studied the stability and magnetic characteristics of cobalt oxide NPs doped with various transition metals, such as V, Cr, Mn, and Fe, shedding light on how different dopants can stabilize magnetic moments in these NPs [20]. Finally, Roy and coworkers investigated the magnetic properties of SnO_2 NPs doped with transition metals through DFT, revealing how doping influences their magnetic behavior [21].

The present work employs the DFT method to investigate the structural and magnetic properties of DyF_3 nanoclusters. By using term "nanoclusters" (NC), we emphasize the limitations of DFT simulations to treat materials with as many atoms in the computational cell as in the real experimental NPs. An important aspect of this study is the application of evolutionary algorithms for the search for the most probable DyF3 NC geometries, enabling the exploration of diverse configurations that might not be accessible through conventional modeling techniques. By analyzing the distribution of magnetic moments as a function of particle size and within the NCs, we seek to advance the fundamental understanding of magnetic phenomena in rare-earth trifluoride NPs and explore their potential applications in various fields. Finally, we address the effect of the presence of solvent on the NC properties.

2. Computational details

The initial structures of DyF_3 NCs were generated using the Universal Structure Predictor: Evolutionary Xtallography (USPEX) code developed by Oganov and coworkers [22–24]. Version 2023.0 of this code was used in the fixed-composition evolutionary algorithm (FCEA) mode to perform structure searches, interfacing with the Vienna Ab initio Simulation Package (VASP, version 6.4.3) for DFT optimization [25] of atomic positions. Structural searches were conducted to identify ground-state structures for NCs with fixed atomic ratios of Dy to F (1 : 3) and varying the number of formula units n from 1 to 10, thus preserving the stoichiometry of the initial structure. Each search consisted of 20 generations, with structures graded based on their enthalpies as calculated by VASP. To accelerate the evolutionary search, simplified computational parameters were used, meaning that the initial screening calculations employed the GGA-PBE functional [26] with a moderate energy cutoff, increased from 350 eV to 450 eV for the first and fifth (final) optimization steps. Only the atomic positions were optimized during these calculations. The convergence criterion for the self-consistent calculations was set to 10^{-4} eV for the initial optimization step and tightened up to 10^{-6} eV for the final step, while all other computational parameters were kept unchanged.

For each composition, the identified ground state structure and several near-degenerate isomers were considered for the subsequent structural relaxation and analysis. Isomers within an energy window of 50 meV relative to the ground state were retained, provided that they represent distinct geometries. Nearly degenerate structures exhibiting only minor distortions were treated as geometrically equivalent and represented by a single representative structure. When reporting size-dependent energetic and magnetic trends, both ground state-specific values and ensemble-averaged quantities were analyzed to distinguish intrinsic structural effects from variations arising due to isomeric diversity.

A cubic simulation box was used for each NC, surrounded by a vacuum region of 15–25 Å in all directions to prevent interaction between periodic images.

Recently, we proposed a methodology for accurately predicting the structural parameters of XF_3 (X = Gd, Tb, Dy, Ho, and Er) trifluorides at 0 GPa [27]. Based on the findings of this research, calculations for DyF_3 NCs were employed within the GGA-PBEsol functional in the spin-polarized mode. The projector augmented wave (PAW) method was employed, and the Dy pseudopotential included $4f^9 5s^2 5p^6 5d^1 6s^2$ electrons in the valence shell with a Hubbard correction [28] of 2.3 eV for $5d$-electrons.

Each structure was re-optimized by the conjugate gradient method until all the force components became less than 0.001 eV/Å, and the convergence condition for self-consistent calculations was the invariant of the total energy of the system with an accuracy of 10^{-8} eV. Brillouin zones were sampled using $1 \times 1 \times 1$ k-points centered at the Gamma point. The Gaussian smearing width was set to 0.05 eV.

The Dy–F bonds were identified within a 3.0 Å cutoff to account for both typical and slightly elongated or distorted bonds, whereas the coordination number (CN) of Dy ions was determined using a stricter 2.7 Å cutoff to include only strong, short-range interactions.

Spin–orbit coupling (SOC) was included into the computational scheme to study the possible features of magnetic anisotropy and calculate the magnetic anisotropy energy (MAE). To do so, the initial spin moment of $8\mu_B$ was imposed on each Dy ion and the MAE was calculated according to the following equation:

$$\mathrm{MAE} = E_{actual} - E_{easy}, \quad (1)$$

where E_{actual} is the energy of the system, when spins are oriented along x, y or z direction, and E_{easy} is the energy of the system in the most favorable direction (easy axis).

Implicit solvation effects were evaluated using VASPsol [29] extension of the VASP code, which implements the polarizable continuum model [30]. The dielectric constant corresponding to water was used. In all solvation calculations, clusters were kept neutral, and no ionic dissociation was assumed, consistent with the treatment of solid-state DyF_3 units in solvent environments.

3. Results

3.1. Structural properties

As a result of USPEX searches, the lowest-energy structural isomer and several higher-lying candidates were selected for each cluster size, n = 1–10, and subsequently re-optimized and analyzed using DFT. Generally, for most cluster sizes, several low-energy isomers were identified within a narrow energy window. Therefore, we distinguish between nearly degenerate isomers that are geometrically equivalent and isomers with comparable energies but qualitatively different structurally.

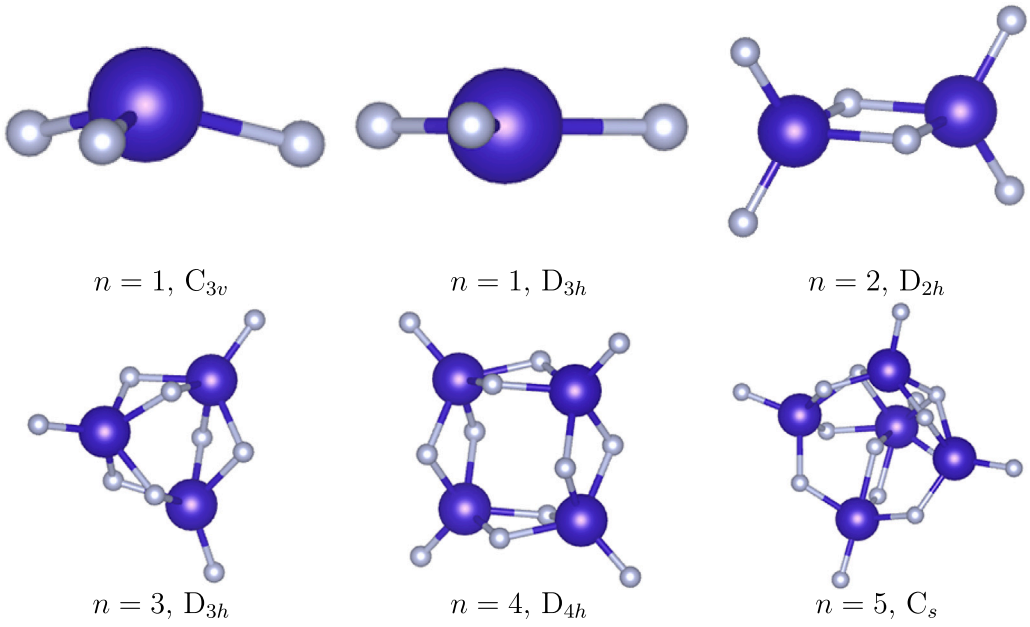


Fig. 1. Optimized $(DyF_3)_n$ nanocluster structures (n = 1–5) obtained by USPEX prediction with corresponding identified point group symmetries.

For small clusters, up to n = 5, multiple low-energy isomers were found within an energy window of 50 meV with respect to the lowest-energy structure with an exception of the $(DyF_3)_1$ monomer, which isomers exhibit very similar structural properties, differing only by minor distortions of DyF_3 building units. In contrast, the monomer displays variations in point-group symmetry even within a narrow energy range below 50 meV. The representative structures discussed here are shown in Fig. 1. As the number of ions in the nanocluster exceeds six, variability in point group symmetries emerges, primarily due to distortions in the lowest-energy structure or rotation of one or several bounds, as seen in Fig. 2, where different isomers of $(DyF_3)_n$ (n = 6–7) are presented in the order of increasing energy.

Overall analysis of the Dy–F, F–F bond and Dy–Dy non-bond lengths, point group symmetries, coordination number of Dy ions, linear size of the clusters, defined as a maximum ion-ion distance, within relative energies, expressed as the differences between the most energetically favorable configuration and a certain isomer are collected in Table 1 (for clusters of nearly degenerate isomers that are geometrically equivalent data is not shown. The complete dataset is provided in Supplementary Materials, Table 1).

Further, structural characteristics will be presented individually for each NC size. For the $(DyF_3)_1$ monomer, all optimized isomers feature a threefold coordinated Dy^{3+} ion. The Dy–F bond lengths in the lowest-energy structure are approximately 2.03 Å, in excellent agreement with the molecular-level results of Xu et al. [31], who reported a C_{3v} pyramidal geometry for isolated DyF_3 with Dy–F distances of 2.023 Å in scalar-relativistic treatment and 2.029 Å when spin–orbit coupling was included in DFT calculations. Our lowest-energy structure S_1^1 (hereafter in S_m^n n is the number of structural units and m is the isomer index), also exhibits C_{3v} symmetry, as well as S_2^1 similarly adopts this symmetry. While Xu et al. [31] identified a C_{3v} pyramidal minimum, they noted that the energy barrier to planarization is especially low for heavier lanthanides such as Dy. Consequently, the vibrationally averaged geometry may resemble D_{3h} symmetry. This behavior is reflected in our S_3^1 structure, which exhibits a symmetric, planar geometry consistent with D_{3h} symmetry as described in their study. The energy difference between this planar geometry and the lowest-energy pyramidal structure is 19.6 meV, further supporting the shallow nature of the potential energy surface in this region.

For all nearly degenerate isomers of $(DyF_3)_2$, the fluoride ions act as bridging ligands between two Dy^{3+} centers, forming a well-defined, symmetric coordination environment. All five isomers exhibit D_{2h} point group symmetry, with each Dy^{3+} ion fourfold coordinated by fluoride ions. The lengths of the Dy–F bonds range from 2.023 to 2.220 Å, while F–F and Dy–Dy distances remain nearly constant at 2.659–2.660 Å and ~3.55 Å, respectively.

For $(DyF_3)_3$ and $(DyF_3)_4$, all low-energy isomers adopt D_{3h} and D_{4h} symmetries, respectively. In $(DyF_3)_3$ configuration, the fluoride ions bridge between Dy^{3+} centers, creating a compact, triangular core. For both NCs, each Dy^{3+} ion is fivefold coordinated by fluoride ligands. Dy–F bonds range from 2.026 to 2.227 Å for n = 3, and 2.022−2.220 Å for n = 4.

In $(DyF_3)_5$ configuration, all five isomers exhibit C_s symmetry. Dy^{3+} ions remain fivefold coordinated, with Dy–F bond lengths ranging from 2.015 to 2.700 Å. Notably, the S_5^5 isomer deviates from this trend, exhibiting Dy–F bonds up to 2.957 Å, suggesting more distorted interactions. The F–F distances span 2.558–2.925 Å, and Dy–Dy distances range from 3.554 to 5.315 Å, reflecting elongation in certain regions of this NC. The first four isomers lie within a narrow 2.9 meV energy window, indicating near-degenerate configurations, whereas S_5^5 is significantly higher in energy (219.3 meV), corresponding to a highly distorted geometry.

The $(DyF_3)_6$ isomers reveal increased structural diversity. The two lowest-energy isomers are near-degenerate and exhibit C_{4v} symmetry and feature Dy^{3+} ions in fivefold coordination with fluoride ligands. Their Dy–F bond lengths span 2.018–2.882 Å, while F–F and Dy–Dy distances range from 2.848–2.958 Å and 3.768–5.727 Å, respectively, indicating a compact and slightly distorted coordination framework. In contrast, the higher-energy isomers deviate more significantly. The S_3^6 and S_4^6 isomers, with C_1 and C_s symmetry, respectively, feature broader Dy–F bond length ranges (up to 2.584–2.612 Å), extended Dy–Dy separations reaching 6.296 Å, and CNs varying between 4 and 6, reflecting less symmetric and more expanded structures. The S_5^6 isomer, although exhibiting idealized O_h symmetry, lies 866.7 meV above the ground state and likely represents a highly strained geometry.

In $(DyF_3)_7$ NCs, all five isomers exhibit low symmetry (C_1, C_s, or C_2). Dy–F bond lengths span 2.009–2.820 Å, F–F distances range from 2.502 to 2.991 Å, and Dy–Dy separations extend from 3.429 to 7.025 Å. CNs vary from 5 to 7. While the first four isomers are relatively low in energy, S_5^7 is 504.9 meV above the ground state, indicating significant structural distortion and metastability.

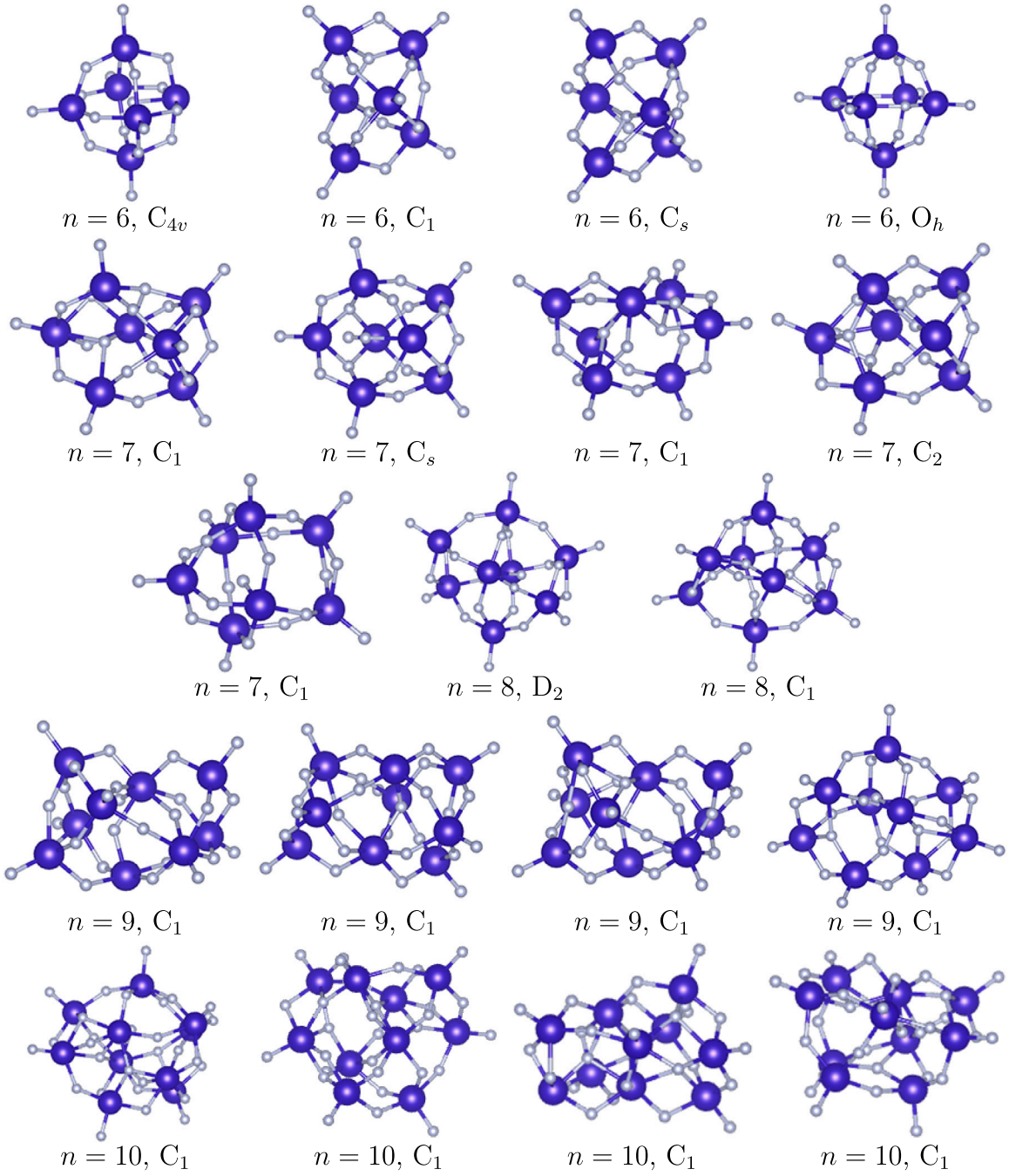


Fig. 2. Different point group symmetries of optimized $(DyF_3)_n$ nanocluster isomers (n = 6–10) obtained from USPEX structure prediction.

The $(DyF_3)_8$ isomers reflect continued geometric variability. The two lowest-energy isomers are near-degenerate and retain D_2 symmetry, with Dy–F bond lengths ranging between 2.018 and 2.804 Å and Dy–Dy distances between 3.512 and 7.429 Å. Higher-energy isomers adopt C_1 symmetry and show slightly greater variations in bond lengths. CNs vary between 5 and 7. Relative energy differences span up to 388.8 meV.

In $(DyF_3)_9$, all isomers exhibit C_1 symmetry. Dy–F distances range from 1.989 to 2.993 Å, F–F from 2.519 to 2.998 Å, and Dy–Dy from 3.568 to 8.372 Å. CNs vary from 5 to 7. The first two isomers are near-degenerate, while the fourth lies 449.7 meV above the ground state.

Finally, $(DyF3)_{10}$ clusters maintain low C_1 symmetry. Dy–F bond lengths range from 2.005 to 2.959 Å, with F–F distances of 2.493–2.995 Å and Dy–Dy separations extending to 8.424 Å. Coordination numbers vary from 5 to 8. The relative energies range up to 809.9 meV.

This systematic analysis reveals that $(DyF_3)_n$ (n = 1–10) NCs evolve from symmetric, well-ordered geometries into increasingly disordered and flexible structures with growing size. Symmetry breaking is accompanied by broader Dy–F and Dy–Dy bond distributions, and variation in CN. Across all sizes, the shortest Dy–F bond lengths remain in the range of 2.015–2.034 Å, slightly shorter than the 2.286 Å Dy–F distance in bulk monoclinic DyF_3 (Materials Project [32–34]), reflecting stronger local bonding in under-coordinated environments. The shortest Dy–Dy separations (3.429–3.768 Å) also stay below the bulk range (3.878 Å),

Table 1
Relative energies, point group symmetries, Dy–F and F–F bond lengths, Dy–Dy distances, Dy coordination numbers, and linear sizes of $(DyF_3)_n$ ($n = 1-10$) nanoclusters.

n[a]	m[b]	E, meV	Sym.[c]	Dy–F, Å	F–F, Å	Dy–Dy, Å	CN_{Dy}[d]	L[e], Å
1	1	0	C_{3v}	2.031–2.032			3	3.492
	3	19.6	D_{3h}	2.034–2.035			3	3.524
2	1	0	D_{2h}	2.023–2.22	2.659	3.557	4	6.652
3	1	0	D_{3h}	2.026–2.227	2.519–2.847	3.43–3.431	5	6.943
4	1	0	D_{4h}	2.022–2.22	2.553–2.999	3.566–5.048	5	9.091
5	1	0	C_s	2.015–2.689	2.559–2.925	3.554–5.293	5	9.285
	5	219.3	C_s	2.016–2.957	2.587–2.974	3.47–5.315	5	9.268
6	1	0	C_{4v}	2.018–2.882	2.848–2.958	3.768–5.727	5	9.764
	3	586.8	C_1	2.011–2.584	2.582–2.981	3.663–6.296	5–6	10.241
	4	663.9	C_s	2.01–2.612	2.708–2.968	3.789–6.156	4–6	10.151
	5	866.7	O_h	2.019–2.211	2.834–2.835	4.152–5.874	5	9.911
7	1	0	C_1	2.01–2.673	2.502 2.985	3.442–6.557	5–7	10.056
	2	82.9	C_s	2.009–2.638	2.587–2.951	3.536–6.282	6	10.09
	3	163.0	C_1	2.019–2.7	2.52–2.991	3.429–7.025	5–7	10.564
	4	392.7	C_2	2.013–2.643	2.615–2.988	3.486–6.296	5–6	9.827
	5	504.9	C_1	2.016–2.82	2.555–2.989	3.555–6.833	5–6	10.575
8	1	0	D_2	2.018–2.804	2.584–2.999	3.512–7.424	5–6	10.988
	3	113.8	C_1	2.014–2.879	2.562–2.993	3.485–7.422	5–7	11.018
	4	388.8	C_1	2.013–2.501	2.518–2.997	3.422–7.589	5–6	11.253
9	1	0	C_1	2.016–2.805	2.557 2.991	3.678 8.257	5–7	11.924
	3	83.0	C_1	1.989–2.582	2.578–2.995	3.571–8.255	5–7	11.937
	4	449.7	C_1	2.002–2.993	2.519–2.975	3.568–8.372	5–6	12.025
10	1	0	C_1	2.008–2.722	2.493–2.979	3.461–8.28	5–7	11.952
	2	279.7	C_1	2.01–2.89	2.496–2.992	3.504–8.29	5–7	12.092
	3	679.0	C_1	2.013–2.959	2.502–2.989	3.441–8.424	5–8	12.237
	4	809.9	C_1	2.005–2.772	2.544–2.995	3.521–7.99	5–8	11.55

[a] Number of structural units.
[b] Index of isomer.
[c] Point group symmetry.
[d] Coordination number of Dy ions.
[e] Linear size of a cluster.

indicating more compact metal substructures. In contrast, Dy^{3+} CNs increase with a number of structural units, from 3 in the monomer to up to 8 in the largest clusters, gradually approaching the ninefold coordination observed in the bulk. F–F distances span 2.493–2.999 Å and closely match the shortest F–F distance in the bulk (2.534 Å). These trends show that although Dy–F and Dy–Dy bonds remain locally compressed, both the coordination environment and overall structure progressively develop bulk-like characteristics with the increase of the NC size.

The size of each NC, defined as the maximum ion–ion distance (L), increases steadily with the number of structural units n, from ≈3.5 Å in the monomer to over 12 Å in $(DyF_3)_{10}$. While this growth is not strictly linear (as seen in SFig. 1, where L is plotted for ground state isomers), it reflects the systematic addition of structural units and the increasing geometric flexibility of the NCs. The broadening of L across isomers at larger sizes also reflects the emergence of diverse, symmetry-lowered configurations.

To investigate the stability of DyF_3 NCs, we calculated the binding energies as $E_b = n(E_{Dy} + 3E_F) - E_n$, where E_{Dy} and E_F are the energies of the isolated Dy and F atoms, respectively, E_n is the energy of a NC with n formula units. Averaged over all generated isomers data is shown in Fig. 3 (a). As expected, the binding energy increases monotonically with a number of structural units, reflecting the cumulative stabilization provided by the growing number of Dy–F interactions. However, relative stability between neighboring cluster sizes is more sensitively captured by the second-order energy difference, $\Delta^2 E = E_{n+1} + E_{n-1} - 2E_n$, evaluated here using the ground-state structures only.

As seen in Fig. 3 (b), second-order energy difference values exhibit a non-monotonic size dependence, with peaks at $n = 3$, 6, and 8, and dips at $n = 2$, 4, 7, and 9. Positive $\Delta^2 E$ indicates enhanced relative stability with respect to adjacent sizes, whereas negative values signal comparatively reduced stability. Importantly, this alternation does not correlate directly with a single local structural descriptor such as Dy coordination number or shortest Dy–F bond length, which vary smoothly across sizes (Table 1).

Inspection of the stablest structures reveal that the addition of DyF_3 units results in compact, internally balanced frameworks without introducing pronounced structural strain. For example, the $n = 3$ cluster adopts a highly symmetric D_{3h} structure with uniform Dy–Dy separations and well-distributed fluoride bridges, while the $n = 6$ and $n = 8$ ground states accommodate additional units primarily by extending existing connectivity motifs rather than creating highly distorted local environments. In contrast, at sizes $n = 2$, 4, and particularly $n = 7$, the ground-state structures show either abrupt changes in global geometry or the emergence of broader Dy–Dy distance distributions and coordination-number variability, indicating structural reorganization rather than smooth growth. Such rearrangements reduce energetic optimality relative to neighboring sizes, resulting in negative $\Delta^2 E$ values.

This behavior highlights that the relative stability of $(DyF_3)_n$ NCs is governed by size-dependent structural evolution rather than simple coordination saturation. A similar conclusion has been reached in recent experimental–theoretical studies of lanthanide halide clusters, where ion-mobility measurements combined with DFT showed that cluster growth proceeds through changes in dominant structural motifs, with relative stability determined by how efficiently a given size accommodates additional units within an evolving three-dimensional framework rather than by local bonding metrics alone [35]. Although the chemical composition and charge state differ, both studies underscore that non-monotonic energetic trends in lanthanide halide clusters originate from global structural reorganization accompanying size growth.

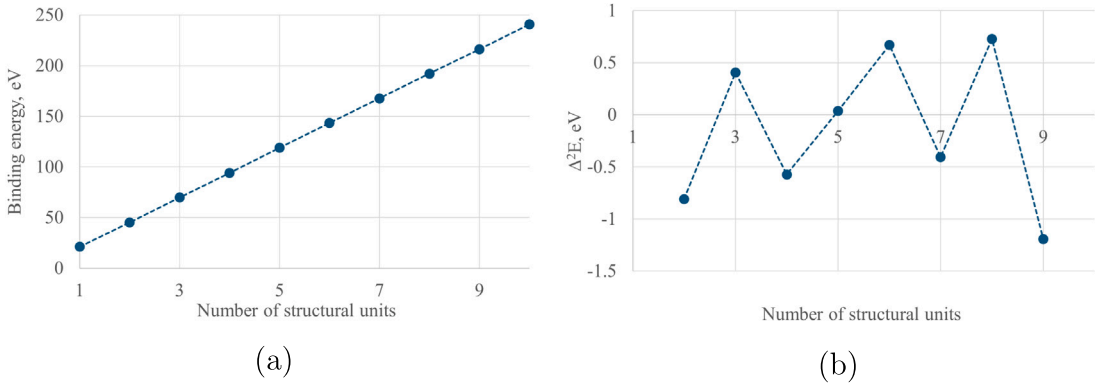


Fig. 3. Binding energy (a) and second-order energy difference (b) dependence on the number of structural units n of $(DyF_3)_n$ nanoclusters.

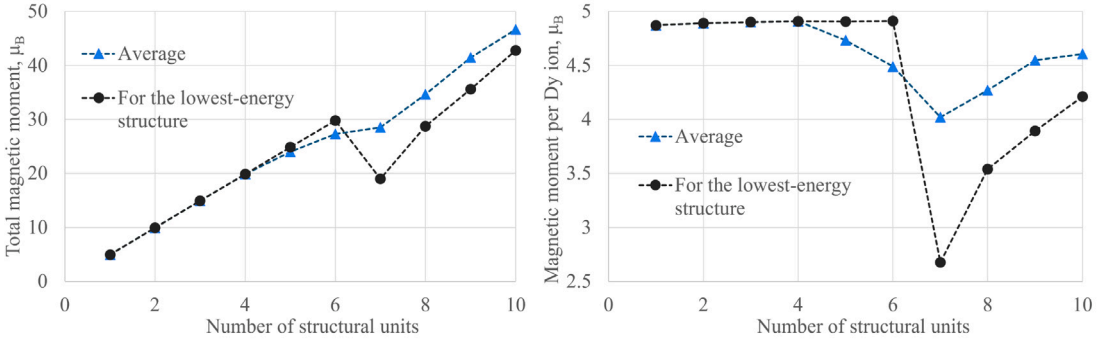


Fig. 4. Total magnetic moment (a) and magnetic moment per Dy ion (b) dependence on the number of structural units n of $(DyF_3)_n$ (n = 1–10) nanoclusters.

3.2. Magnetic properties

3.2.1. Spin-polarized calculations

At the next stage, the magnetic properties of the $(DyF_3)_n$ NCs were analyzed. Total magnetic moments and magnetic moments per Dy^{3+} ion were calculated for each isomer. For each number of structural units, we analyzed both the ground state structure and the full set of isomers by averaging over all available configurations. The resulting trends are presented in Fig. 4, which includes both mean values and those corresponding to the lowest-energy structures.

It follows from the plots that the total magnetic moment of the $(DyF_3)_n$ NCs increases with n (Fig. 4 a), being consistent with the growing number of Dy^{3+} ions. The averaged magnetic moment values rise nearly linearly, while ground state structures follow a similar trend for small sizes but deviate significantly at n = 7–10, where the total moments are markedly reduced, indicating a significant change in spin alignment.

The average magnetic moment per Dy ion shows a more complex, non-monotonic behavior (Fig. 4 b). For small NCs (n = 1–4), the per-ion moments are nearly constant, in the range of 4.87–4.91μ_B, indicating well-isolated Dy^{3+} ions with minimal magnetic coupling. At n = 2, the average value (4.892μ_B) matches exactly our previous bulk calculation [27], and remains close to the reference DFT value of 4.984μ_B reported by Saini [36]. The lowest-energy structures at these sizes show nearly identical values, confirming the uniformity of local Dy spin moments.

At intermediate numbers of structural units (n = 5–7), a noticeable reduction of the average Dy^{3+} magnetic moment is observed, with a minimum of 4.02μ_B at n = 7. In the ground state structures, this suppression is even more pronounced, reaching as low as 2.68μ_B per Dy ion at n = 7. This trend reflects the emergence of strong Dy–Dy magnetic interactions and magnetic frustration, likely driven by decreased local symmetry and complex spin arrangements.

The magnetic behavior of the $(DyF_3)_n$ NCs is governed primarily by the localized 4f electrons of Dy^{3+} ions. Due to the strong localization of the 4f orbitals, direct exchange interactions between Dy ions are negligible. Instead, magnetic coupling is mediated indirectly through Dy–F–Dy super-exchange pathways involving hybridization with ligand p states. Such ligand-mediated super-exchange interactions are well established within generalized super-exchange theory for lanthanide pairs [37] and can give rise to antiferromagnetic or magnetically compensated configurations, particularly in low-symmetry and geometrically frustrated environments. In finite clusters, reduced coordination numbers and distorted Dy–F–Dy angles further enhance the competition between different exchange pathways, leading to non-collinear spin arrangements and partial or complete cancellation of local magnetic moments.

Ion-resolved magnetic data (presented in Table 2 of Supplementary Materials) analysis reveals strongly suppressed or even negative spin moments at certain Dy sites in some isomers, further supporting the presence of non-collinear or antiparallel spin arrangements. Representative structures exhibiting such behavior are shown in Fig. 5. This behavior suggests the appearance of geometric frustration. Moreover, the lowest-energy structures often do not correspond to the magnetically most aligned configurations. Instead, the energetic stabilization of the ground state is associated with magnetic compensation effects, particularly evident at n = 7, where the ground state exhibits the lowest per-ion magnetic moment across the series.

For larger clusters (n = 8–10), the per-ion magnetic moments gradually increase, suggesting partial recovery of stable local environments and spin alignment more consistent with bulk-like magnetic behavior. Nevertheless, the continued discrepancy between ground state and averaged values indicates that magnetic frustration persists even in these larger systems.

In all cases, the calculated magnetic moments remain below the experimental value of 8 μ_B per Dy^{3+} [38] in the bulk crystal.

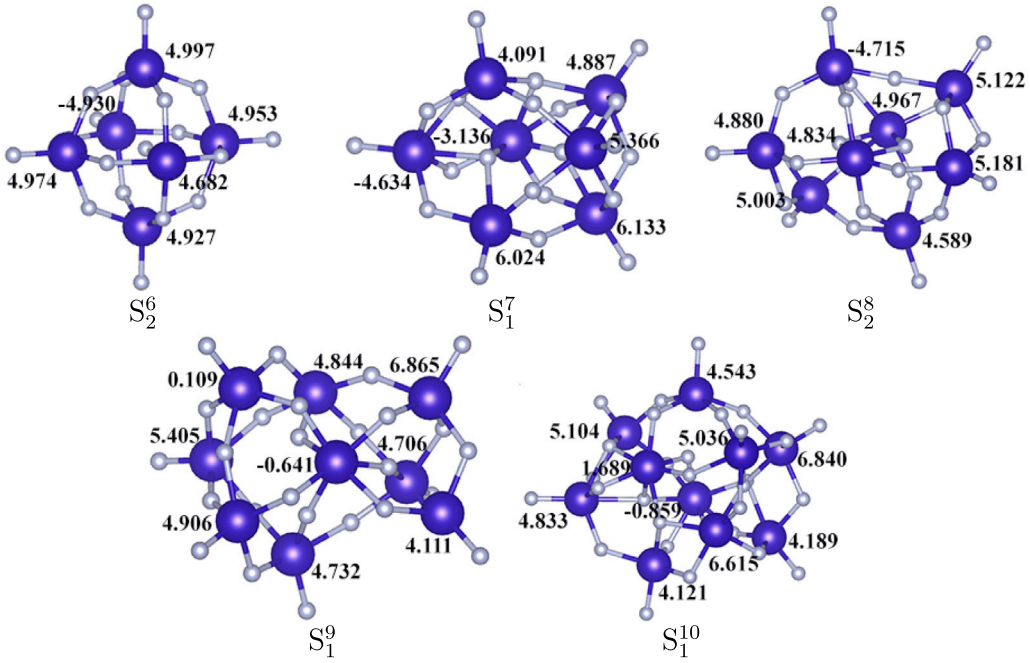


Fig. 5. Distribution of Dy atomic spins for selected $(DyF_3)_n$ nanocluster isomers (n = 6–10).

Furthermore, SOC inclusion in the computation scheme did not significantly influence neither overall magnetic moments nor moment per Dy ion spins. However, SOC is important when studying magnetic anisotropy.

3.2.2. Magnetic anisotropy

The magnetic anisotropy energy (MAE) is a critical parameter for characterizing magnetic materials, as it defines the energy barrier for reorienting the magnetization direction. In this section, the MAE of the most stable DyF_3-based NCs was evaluated to understand how subtle geometric and electronic differences influence their magnetic behavior.

For MAE calculations, spin orientations along three mutually perpendicular directions were considered. The resulting energies were calculated according to Eq. (1) and presented in Table 3 of Supplementary Materials. It is important to note that the x, y, and z labels, used for spin orientation, do not correspond to fixed Cartesian axes. Instead, they refer to three orthogonal directions that are approximately aligned with the structural geometry of each cluster. Given the quasi-symmetric shapes of most NCs, this approach enables a consistent and meaningful comparison of magnetic anisotropy.

Schematic representation of the distribution of easy and hard axes of the NCs is shown in Fig. 6. As seen from it, all NCs exhibit significant magnetic anisotropy, with a distinct preference for spin alignment along one or two directions. In some clusters, however, two directions display nearly identical energies, suggesting the existence of two nearly degenerate easy axes. The remaining orientations are associated with higher energies and can be interpreted as hard axes. The magnitude of MAE varies significantly with cluster size, reaching up to 245 meV for n = 10, which points to a strong uniaxial anisotropy.

Overall, the results confirm that DyF_3 NCs possess strong and size-dependent anisotropy, making them promising candidates for applications in spintronics and quantum information storage.

3.3. Solvent effect

To evaluate the realistic stability of $(DyF_3)_n$ NCs, solvation effects were incorporated into the computational scheme, as these systems are typically synthesized, processed, or applied in aqueous environments. In such media, solvent interactions can strongly influence structural and thermodynamic behavior, especially for ionic compounds like rare-earth fluorides. Solvent screening reduces electrostatic repulsion and stabilizes the undercoordinated sites, potentially altering the relative stability of gas-phase isomers. Therefore, including solvation is essential to link computational predictions with experimental conditions.

Solvation energy quantifies the energy change when a solute is transferred from a vacuum into a solvent environment. It typically reflects the enthalpy change associated with the stabilization of the solute by the surrounding dielectric medium.

In this study, solvation energies were evaluated for each number of structural units and across all available isomers as the energy difference between the solvated and vacuum systems as $E_{solv} = E_{total}^{solvated} - E_{total}^{vacuum}$. Furthermore, both isomer-resolved and average values were analyzed. The calculated mean solvation energies for $(DyF_3)_n$ nanoclusters (n = 1–10) are presented in Fig. 7, while isomer-resolved values are collected in Table 2 of Supplementary Materials. As seen, the mean solvation energy shows a consistent trend of increasing stabilization with the number of structural units, ranging from −2.81 eV for the monomer to −8.17 eV for the $(DyF_3)_{10}$ nanocluster. This monotonic decrease indicates enhanced solute–solvent interaction as the NC grows, likely due to increased surface area and greater charge delocalization in larger clusters. Notably, the most stabilized individual isomers do not always correspond to the lowest-energy gas-phase structures. For example, among the $(DyF_3)_7$ isomers, E_{solv} ranges from −6.00 eV to −6.60 eV, with the most stabilized structure S_7^4 not being the global minimum in vacuum. These findings suggest that the solvent environment can shift the relative energetic ordering of isomers compared to vacuum, emphasizing the need to account for solvation effects when modeling the structure and thermodynamic behavior of lanthanide-based NCs in solution.

4. Conclusions

This work presents a systematic DFT study of the structural, magnetic, and solvation properties of $(DyF_3)_n$ NCs (n = 1–10), revealing a rich and size-dependent evolution of behavior across the NCs sizes.

Structurally, small clusters ($n \leq 4$) exhibit compact geometries with higher symmetries and strong local bonding, where Dy ions show lower coordination numbers and shorter Dy–F and Dy–Dy distances compared

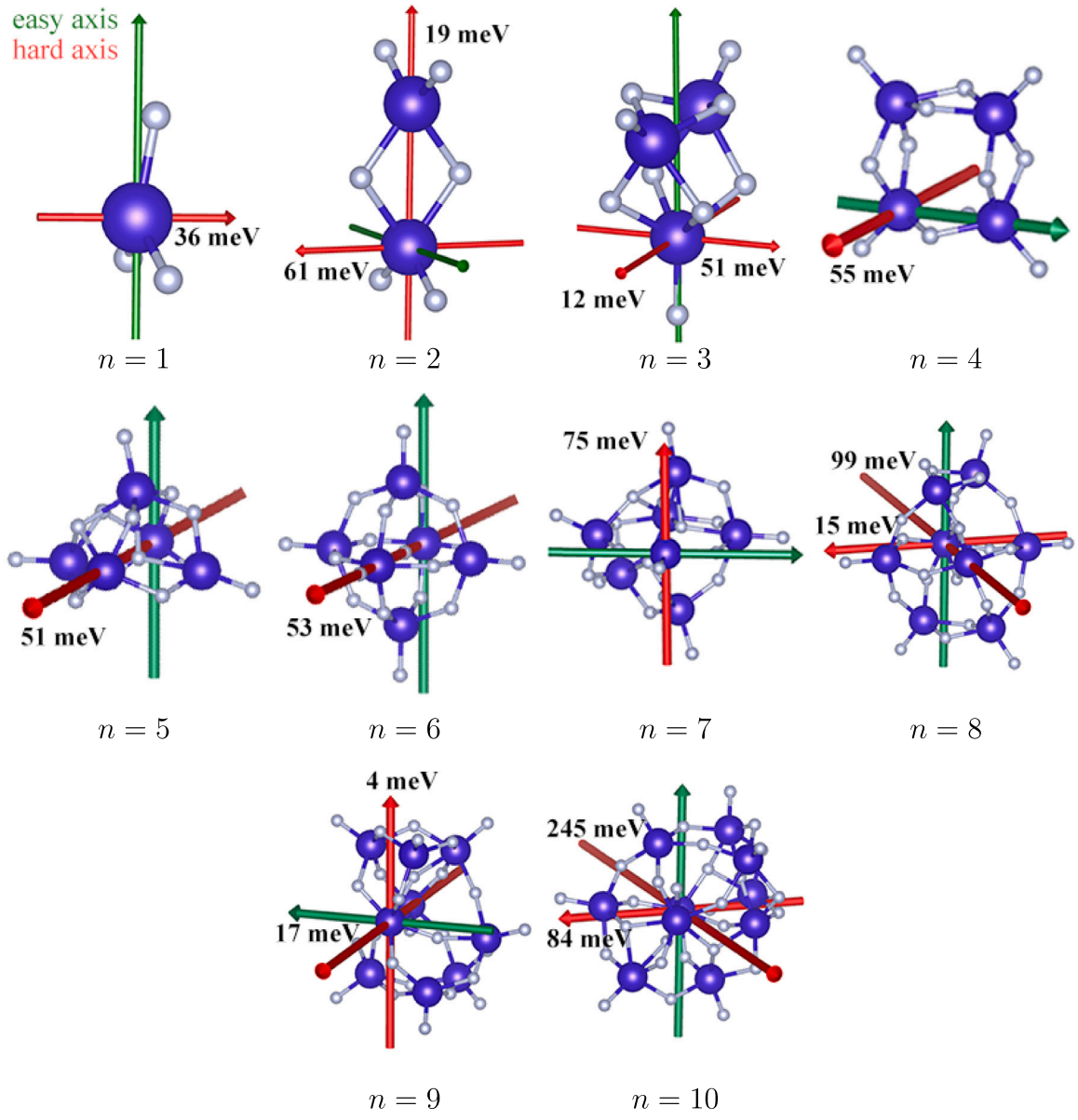


Fig. 6. Schematic representation of easy and hard axes of the most stable $(DyF_3)_n$ (n = 1–10) nanoclusters and corresponding magnetic anisotropy energies.

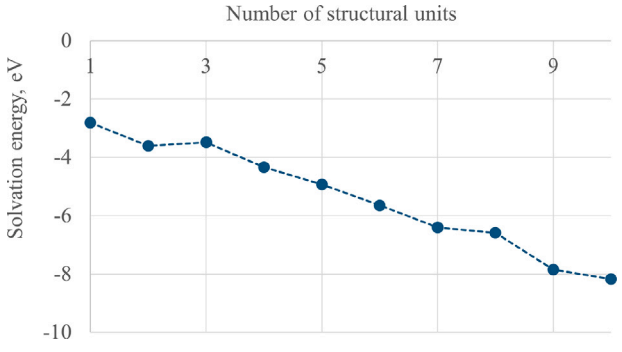


Fig. 7. Solvation energy of $(DyF_3)_n$ (n = 1–10) nanoclusters.

to the bulk material. As the number of formula units grows, the NCs progressively lose symmetry and become more geometrically diverse and flexible.

Binding energy analysis confirms increasing thermodynamic stability with a number of structural units, while the second-order energy difference highlights a number of structural units (n = 3, 6, 8) associated with particularly favorable configurations.

Magnetic properties also exhibit size-dependent behavior. In small NCs (n = 1–4), Dy^{3+} ions behave as nearly isolated spins with uniform magnetic moments. At intermediate n = 5–7, reduced average moments and non-trivial spin alignments indicate the onset of inter-ion interactions and magnetic frustration. Larger NCs (n = 8–10) begin to recover bulk-like alignment, although the calculated magnetic moments remain lower than experimental values, probably due to missing orbital contributions. Importantly, the magnetically most aligned isomers are not always the most stable, indicating a compromise between structural and magnetic energy minimization.

In addition to overall magnetic behavior, all NCs exhibit strong magnetic anisotropy, with the preferred spin orientation. Most clusters show a distinct energy minimum for one or two orientations, indicating pronounced anisotropic behavior that evolves with structure. The magnitude of the anisotropy energy increases with cluster size, reaching

up to 245 meV at $n = 10$. This confirms that DyF_3 NCs possess size-dependent uniaxial anisotropy, reinforcing their potential for spintronic and quantum information applications.

The calculated solvation energies show a consistent trend of increasing stabilization with cluster size, driven by enhanced solvent interaction with larger surface areas and charge delocalization. Notably, the most stabilized solvated isomers do not always correspond to vacuum ground states, suggesting that solvent effects can alter relative stabilities and potentially favor alternative structures in solution. This highlights the importance of including solvent interactions in the realistic modeling of lanthanide NCs.

Overall, these findings shed light on the design and application of lanthanide fluoride nanomaterials, especially in fields where size-dependent properties are critical. In particular, the DyF_3 nanoclusters studied here represent well-defined model systems to probe how structural evolution and magnetic behavior are modulated at the nanoscale. Their progressive stabilization in solvent and sensitivity to local geometry suggest potential for fine-tuning properties relevant to biomedical, spintronic and quantum information applications such as magnetic resonance imaging, where both solution behavior and magnetic response are crucial.

CRediT authorship contribution statement

Regina M. Burganova: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Zafari Umar:** Writing – review & editing, Methodology, Investigation. **Ilya V. Chepkasov:** Writing – review & editing, Validation, Resources, Methodology, Investigation. **Hayk Zakaryan:** Writing – review & editing, Validation, Supervision, Project administration. **Irina I. Piyanzina:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors have declared no conflict of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.comptc.2026.115657.

Data availability

Data will be made available on request.

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